

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 11)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

1. Write a Python program (named `Lab11_1.py`) with 2 functions
 - Write a function `countIntegers()` with *arbitrary parameters*, which:
 - Accepts *parameters*: (`para1`, `para2`, `*paraMore`)
 - The function body check, count, and display the total number of `int` type arguments passed.
 - *No return* value is needed.
 - Write the function `main()` to test the function above, with the sample code below

Sample Program Code, reference only

```
# Lab11_1.py, ICP, 2021

def countIntegers(para1, para2, *paraMore):
    # .. and MORE
    print("Total number of integer arguments is", totalInt)

def main(): # the main function
    countIntegers(20214321, "CHAN Tai Man")
    countIntegers(20214020, "CHAN Siu Ming", [100,90,99,100])
    countIntegers(20214020, "CHAN Siu Ming", 100, 90, 99, 100)

main() # start execution
```

Sample Program Output

```
Total number of integer arguments is 1
Total number of integer arguments is 1
Total number of integer arguments is 5
```

2. Write a Python program (named **Lab11_2.py**) with 2 functions

- Write a *recursive function* **dispRecur()**, which:
 - Accepts 1 parameter (a List expected): (**aList**)
 - The function body simply check the input parameter.
 - If input parameter is empty (an empty list), the function terminates.
 - If input parameter is not empty
 - It first displays the input parameter;
 - It then calls the function itself recursively, with an updated list which removes the first element with `del(aList[0])`
 - *No return* value is needed.
- Write the function **main()** to test the function above, with the sample code below

Sample Program Code, reference only

```
# Lab11_2.py, ICP, 2021

def dispRecur(aList):
# .. and MORE

def main(): # the main function
    dispRecur( [20214020, "CHAN Siu Ming", 100, 90, 99, 100] )
    print()
    dispRecur( list("*****") )

main() # start execution
```

Sample Program Output

```
[20214020, 'CHAN Siu Ming', 100, 90, 99, 100]
['CHAN Siu Ming', 100, 90, 99, 100]
[100, 90, 99, 100]
[90, 99, 100]
[99, 100]
[100]

['*', '*', '*', '*', '*', '*', '*', '*', '*', '*']
['*', '*', '*', '*', '*', '*', '*', '*', '*']
['*', '*', '*', '*', '*', '*', '*', '*']
['*', '*', '*', '*', '*', '*']
['*', '*', '*', '*', '*']
['*', '*', '*', '*']
['*', '*', '*']
['*', '*']
['*']
```